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ASSIGNMENT 2

CRITICAL REVIEW OF EACH SECTION

Paper Reviewed: *A Survey of Vulnerabilities in Federated Learning (2021)*
Type: Section-wise Critical Analysis

1. Abstract — Critical Review

The abstract effectively introduces the theme of the paper, which is the security and privacy risks in Federated Learning (FL). It clearly indicates that the paper focuses on vulnerabilities, attacks, and potential defense strategies. The reader quickly understands the research direction without needing to read the full paper.

Strengths:
The abstract is concise and well-organized. It clearly states the topic, importance, and scope of the study. It also informs the reader that the paper is a survey, not an experiment-based study.

Weaknesses:
The abstract does not mention any major findings or key conclusions. It would be stronger if it briefly stated which vulnerability is most harmful or which defense strategy seems promising.

Evaluation:
Overall, the abstract communicates the topic successfully but lacks analytical insight into results or outcomes.

2. Introduction — Critical Review

The introduction explains Federated Learning, its motivation, and why privacy is important in modern machine learning systems. It explains that FL helps protect data by avoiding centralized storage.

Strengths:
The section gives a strong background for beginners. It clearly differentiates FL from traditional machine learning. The discussion of privacy is relevant and well-timed.

Weaknesses:
The introduction focuses more on explanation than analysis. It does not compare existing studies or identify gaps in current research. There is also limited critical evaluation of why earlier security methods may fail in FL environments.

Evaluation:
The introduction sets context well but lacks research motivation and analytical depth.

3. System Model and Background — Critical Review

This section describes the architecture of FL, including how clients and servers interact and how training occurs without data sharing.

Strengths:
The system model is clearly defined. It helps the reader visualize FL architecture and understand data flow.

Weaknesses:
There is little emphasis on how the architecture itself increases risk. For example, distributed updates and untrusted clients are not analyzed deeply as security challenges.

Evaluation:
Technically clear but lacks security-focused discussion.

4. Threat Model — Critical Review

The threat model identifies different types of attackers such as malicious clients and compromised servers.

Strengths:
The section considers both insider and outsider attacks. It also assumes realistic adversarial capabilities.

Weaknesses:
The threats are not prioritized by severity. No discussion is provided on likelihood or real-world threat frequency.

Evaluation:
Good theoretical structure, but weak risk evaluation.

5. Vulnerabilities in Federated Learning — Critical Review

This section describes weaknesses like information leakage, model inversion, and system exploitation.

Strengths:
This is one of the strongest sections. Vulnerabilities are explained clearly and in technical detail.

Weaknesses:
The impact of vulnerabilities is not measured. For example, accuracy loss, model corruption, or privacy loss are not quantified.

Evaluation:
Strong explanation, weak impact analysis.

6. Attacks in Federated Learning — Critical Review

The paper discusses poisoning, inference, and manipulation attacks.

Strengths:
The attack types are well categorized. Each attack is logically presented and understandable.

Weaknesses:
There is no experimental validation. The success rate and complexity of attacks are not investigated.

Evaluation:
Informative but lacks empirical support.

7. Defenses and Mitigation Techniques — Critical Review

The paper covers methods like secure aggregation and differential privacy.

Strengths:
Important defense approaches are included. The authors correctly identify privacy and integrity goals.

Weaknesses:
No performance cost is discussed. There are no results to prove which defense is more effective.

Evaluation:
Conceptually strong but technically incomplete.

8. Future Research Directions — Critical Review

This section highlights gaps in current research.

Strengths:
Encourages further work and recognizes unsolved issues.

Weaknesses:
Suggestions are general and unspecific.

Evaluation:
Helpful but shallow.

9. Conclusion — Critical Review

The conclusion summarizes the main concerns of FL security.

Strengths:
Clear recap.

Weaknesses:
No actionable recommendations.

Evaluation:
Good summary but weak closing analysis.

OVERALL EVALUATION

Area	**Evaluation**
Technical Explanation	Excellent
Practical Implementation	Weak
Critical Analysis	Moderate
Experimental Evidence	Missing
Industry Relevance	Low

FINAL ASSESSMENT / CONCLUSION

The paper is a strong theoretical survey but weak in practical validation. It is suitable for academic understanding but insufficient for system design.

This paper succeeds in explaining security problems in Federated Learning but does not experimentally test solutions. It is useful as a learning resource, not as a deployment guide.